Spreading / overlap relief

Spreading pushes overlapping or near pairs apart until a minimum pairwise distance holds

The idea

- While any pair sits closer than minDist, push them along their separation vector
- Finish with a deterministic repair pass so the result is stable
- Spreading is a legality proxy, not full row-site legalization

Pseudocode

- Spread lite pushes close pairs apart until every pairwise distance clears a minimum
- It is overlap relief, not site legalization
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Start from the triple-overlap demo, run the spreader with minimum distance zero point five

Algorithm sketch — try these

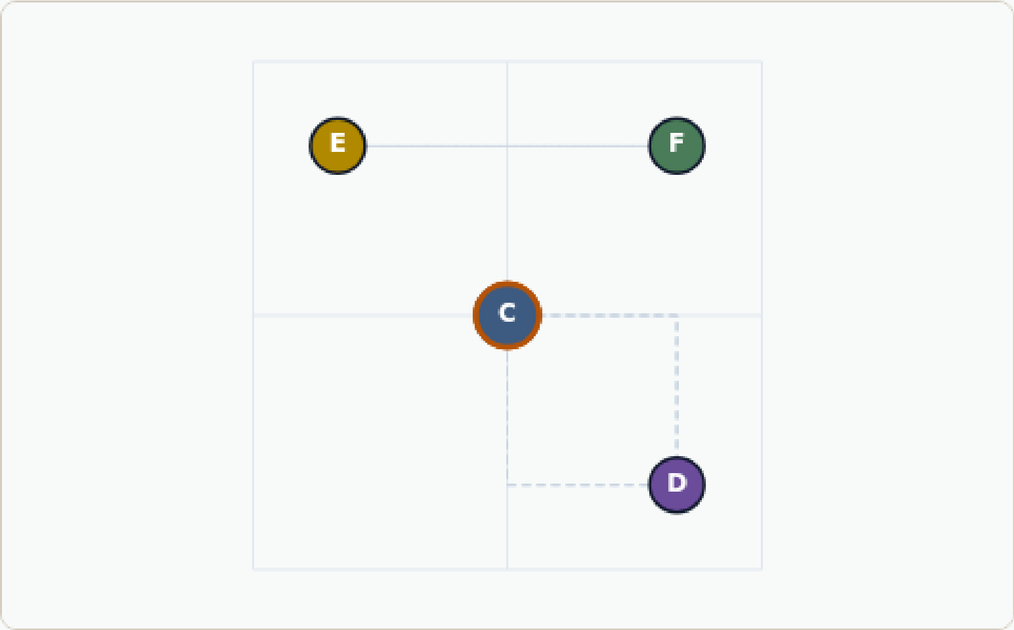
```
INPUT: positions, min_dist
OUTPUT: spread positions
while exists pair with dist < min_dist:
  push the pair apart along their vector
stop when all pairs clear min_dist
NOTE: not row/site legalization
GOLDEN min_dist=0.5 on overlap seed
```

Triple overlap at one point

SPREAD / LEGALIZE LITE

Triple overlap at one point

Step 1 / 5 · triple-overlap · module03-03-spread-legalize-lite



Explanation

The overlap demo stacks A, B, and C on (4,4). Min pairwise distance is zero—illegal for any site-aware flow, perfect for a spreading lesson.

- A,B,C @ (4,4)
- minPairDist = 0
- D,E,F already spread

```
minPairDist: 0
Target minDist: 0.5
```

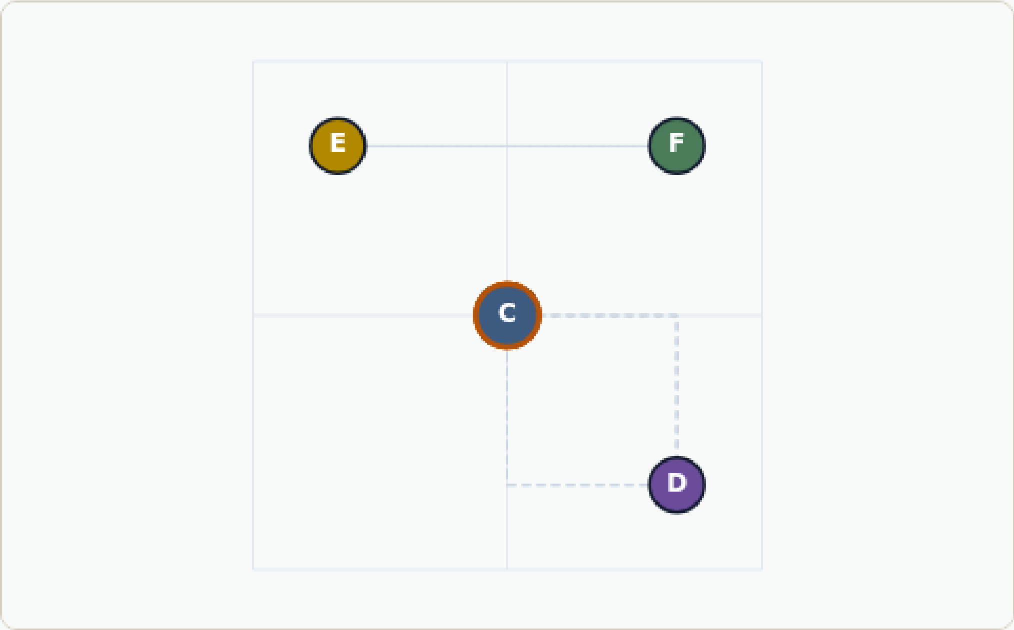
Triple overlap at one point

Push near pairs apart

SPREAD / LEGALIZE LITE

Push near pairs apart

Step 2 / 5 · push-apart · module03-03-spread-legalize-lite



Explanation

While any pair sits closer than minDist, push them along their separation vector. Repeated passes peel the triple stack into distinct points.

- Push along separation vector
- strength controls step size
- Not full row-site legalization

```
minDist = 0.5
iters ≈ 40 + repair
```

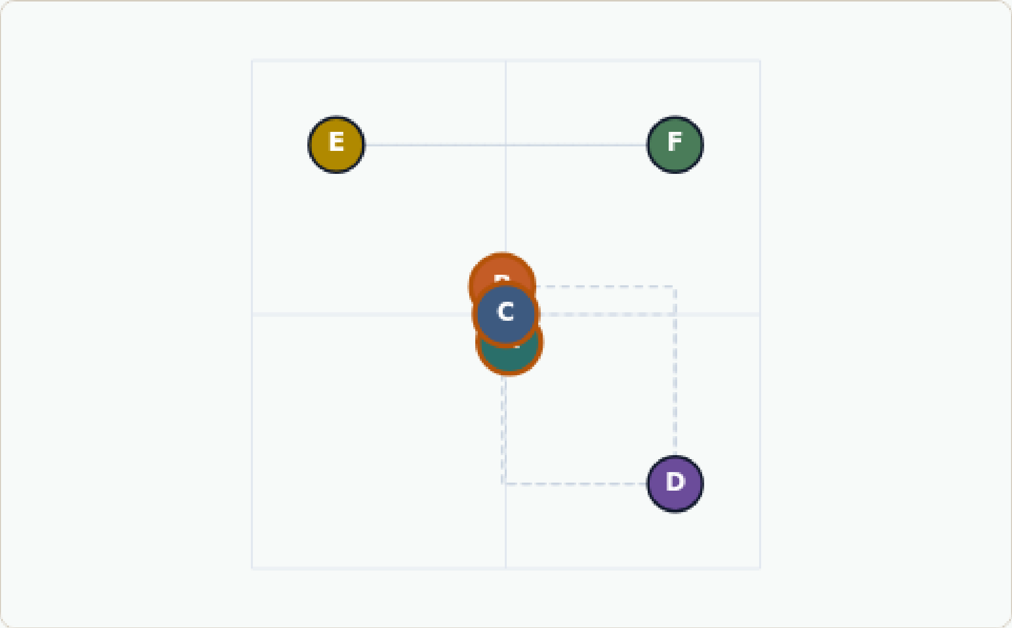
Push near pairs apart

After spread: min distance holds

SPREAD / LEGALIZE LITE

After spread: min distance holds

Step 3 / 5 · after-spread · module03-03-spread-legalize-lite



Explanation

The lite spreader separates A, B, and C until every pair clears about zero point five. D, E, and F barely move.

- minPairDist ≥ 0.5
- Deterministic repair pass
- HPWL may rise—that is expected

```
minPairDist: 0.5
spreadMinPairDist golden: 0.5
```

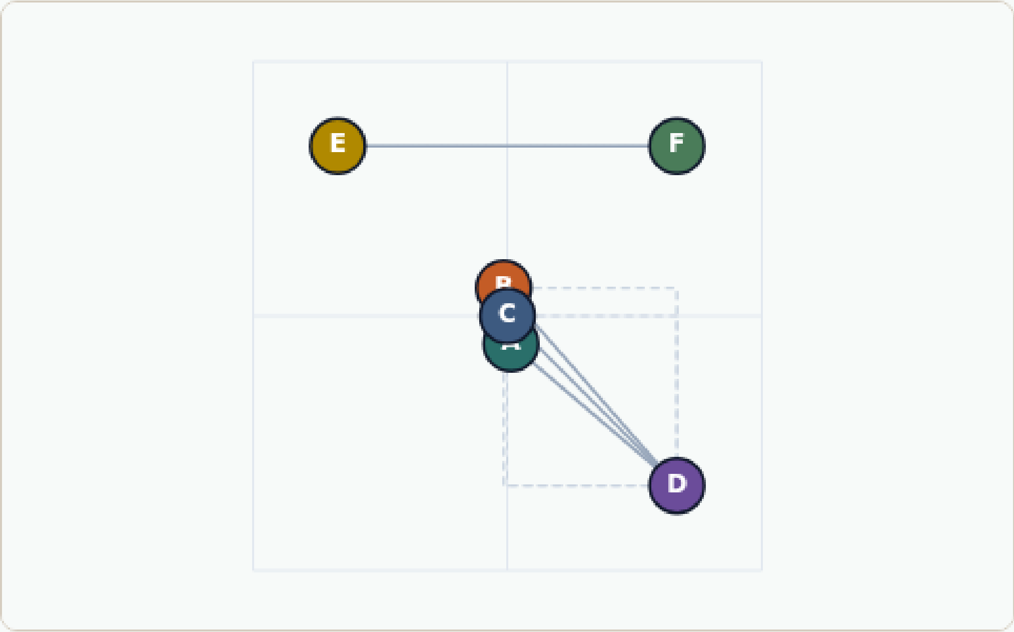
After spread: min distance holds

Spreading is a legality proxy

SPREAD / LEGALIZE LITE

Spreading is a legality proxy

Step 4 / 5 · legality-proxy · module03-03-spread-legalize-lite



Explanation

Clearing min distance is not row legalization or site snapping. It is a teaching stand-in so overlap stops hiding behind pretty HPWL.

- Proxy ≠ detailed placement
- Still report HPWL after
- Use after analytical collapse

```
HPWL after spread: 26.8
minDist: 0.5
```

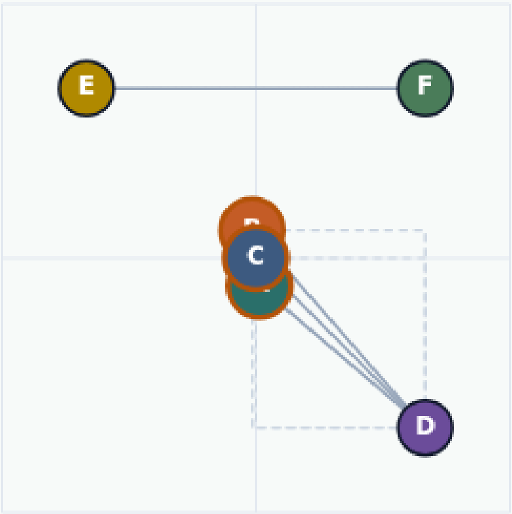
Spreading is a legality proxy

Relieve overlap before celebrating WL

SPREAD / LEGALIZE LITE

Relieve overlap before celebrating WL

Step 5 / 5 · takeaway · module03-03-spread-legalize-lite



Explanation

Start from the triple-overlap seed, spread to minDist zero point five, and confirm every pair clears the threshold. Then revisit wirelength.

- Overlap seed → spread
- minDist 0.5 golden
- Next: timing-driven weights

```
spreadMinPairDist: 0.5
```

Relieve overlap before celebrating WL

Browser lab track

- In the browser lab track, open the **spread-legalize-lite** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track
- Parse `tiny_place.json`, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you’re ready, take the short quiz, then continue to the next module